DEVELOPMENT OF FACE AND PALATE

PREPARED BY:

UMM-E-AYMAN AKBAR

BDS 1ST YEAR

BATCH V

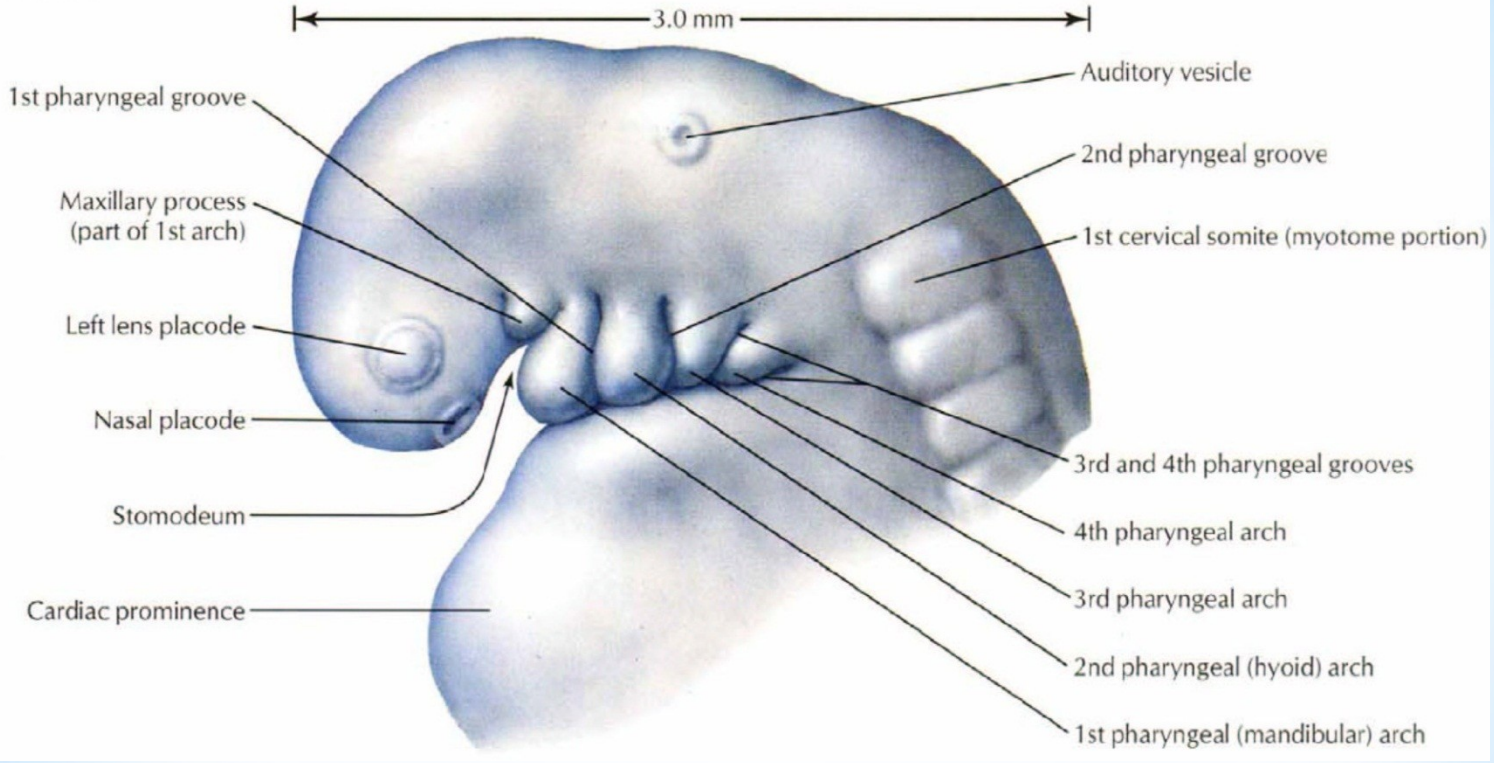
DIDC,DUHS

That duck face you keep making in all of your pictures? Yeah, it's as ugly as sin. If God wanted you to be a duck, you'd have been quacking a long time ago.

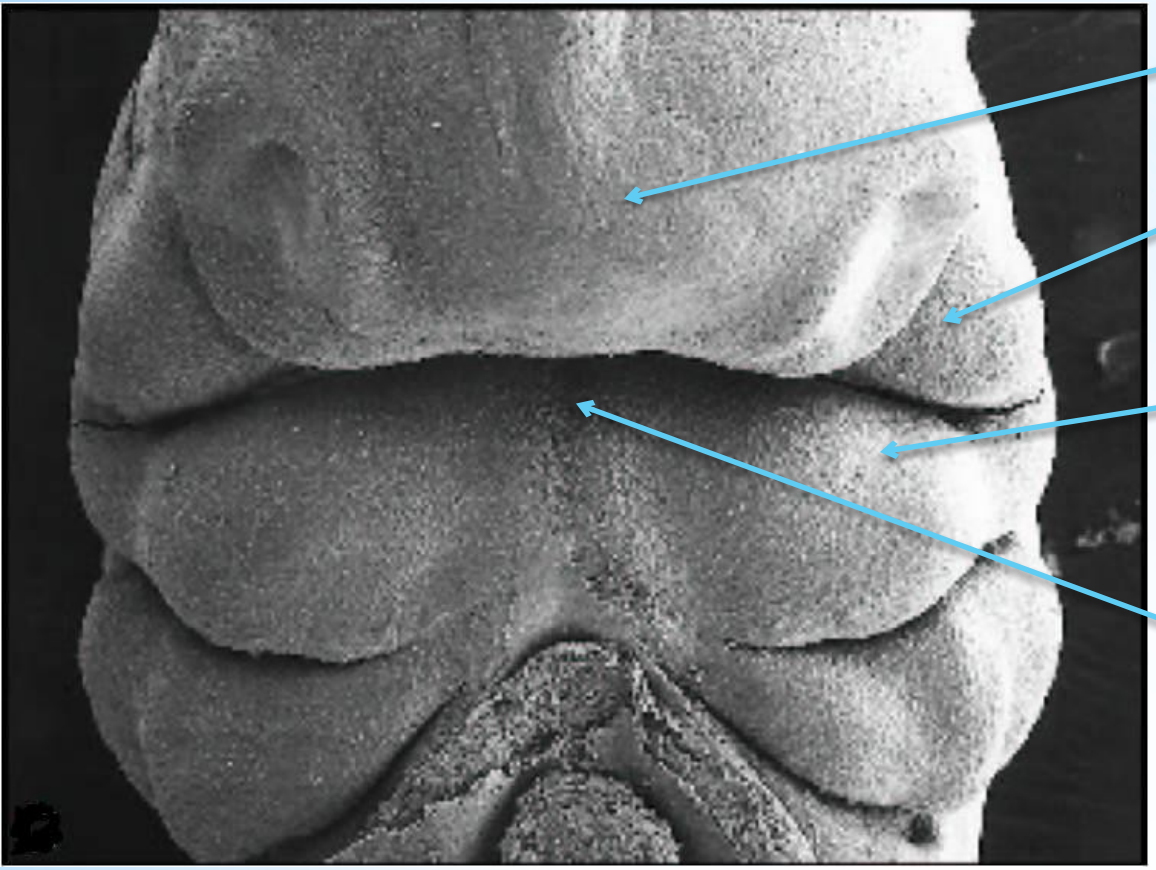




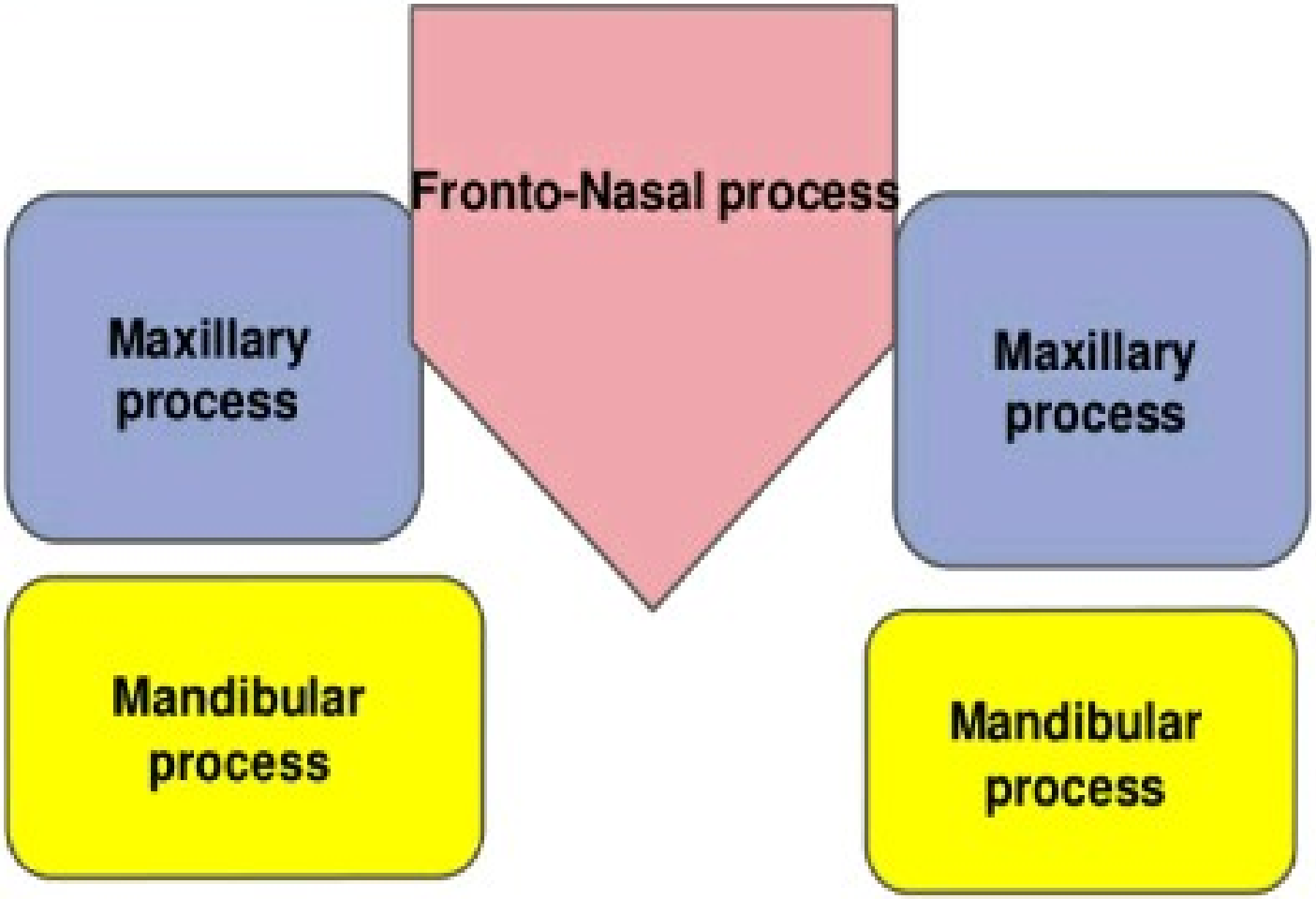




* The development of face occurs mainly between 4th – 8th week.
* Early in the 4th week, five primordial swellings appear around the stomodeum and play an important role in the development of face.
* While the facial proportions develop during the fetal period (9th week to birth).

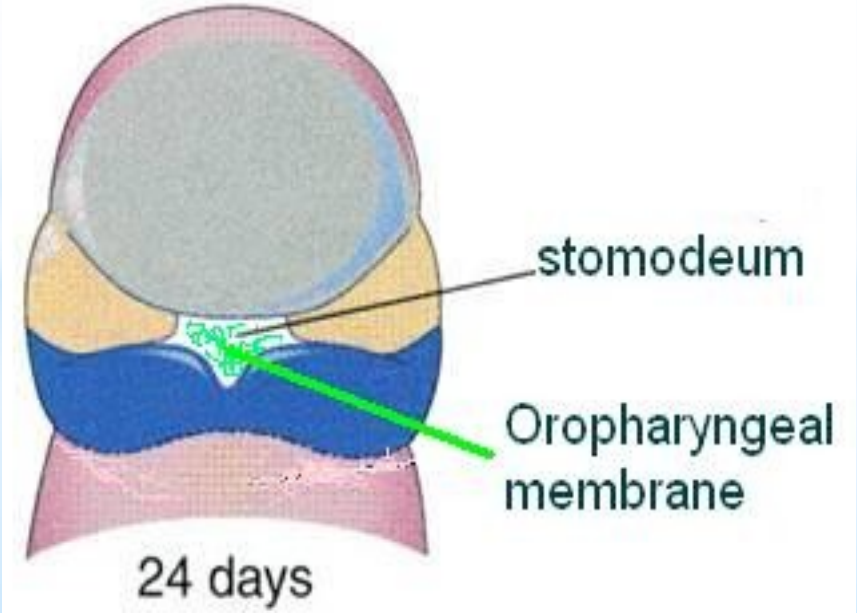

Frontonasal prominence
Maxillary prominences
Mandibular prominences
Stomodeum

DEVELOPMENT OF FACE

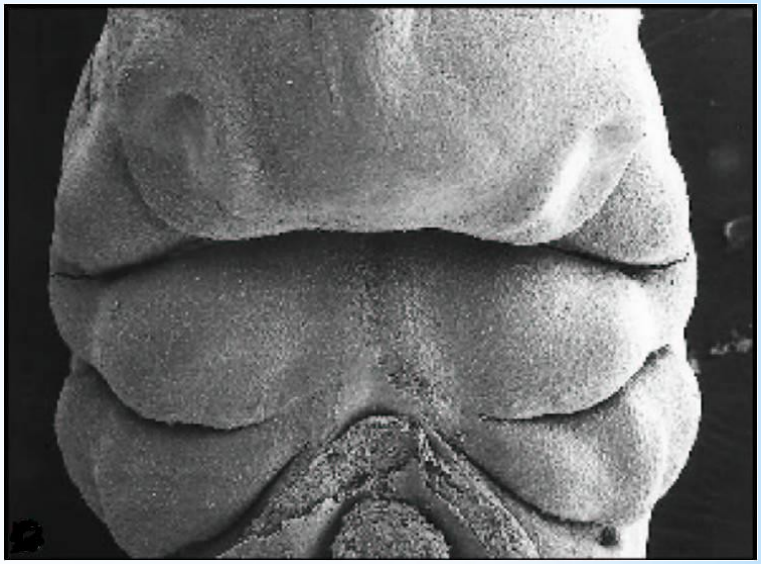


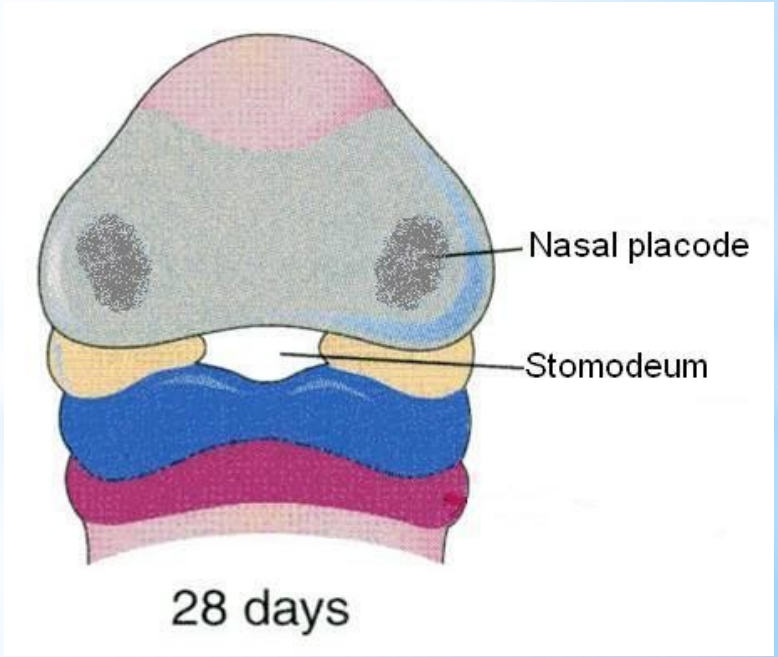
STOMODEUM

- An ECTODERM line depression.
- Separated from the primitive pharynx by the buccopharyngeal (oropharyngeal) membrane.
- The membrane later breaks down and stomodeum opens into the pharynx.

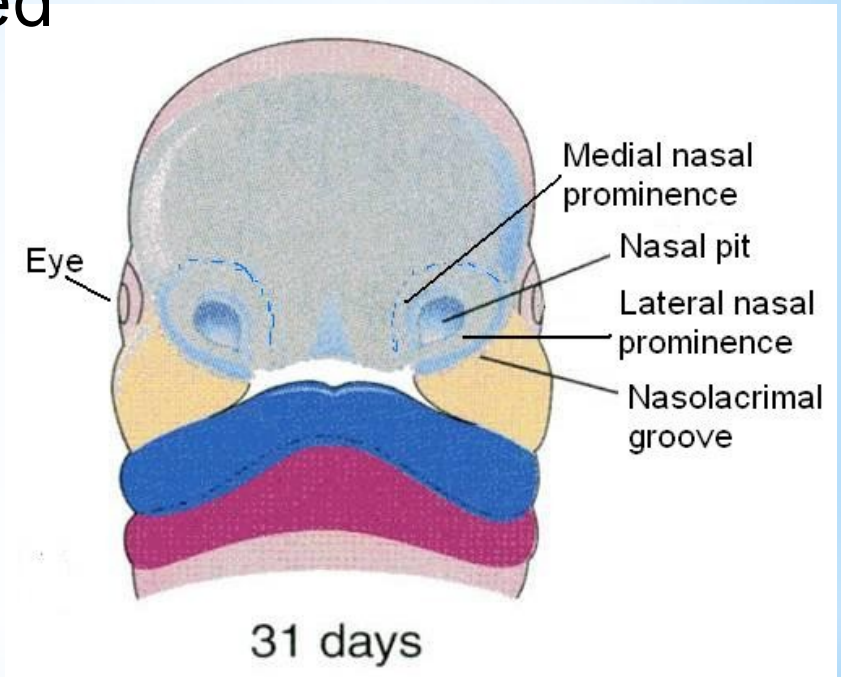


- By the end of 4th week, bilateral oval-shaped ectodermal thickenings called ‘nasal placodes’ appear on each side of the lower part of the frontonasal prominence.
- Nasal placodes are primordia of the nose and nasal cavities.



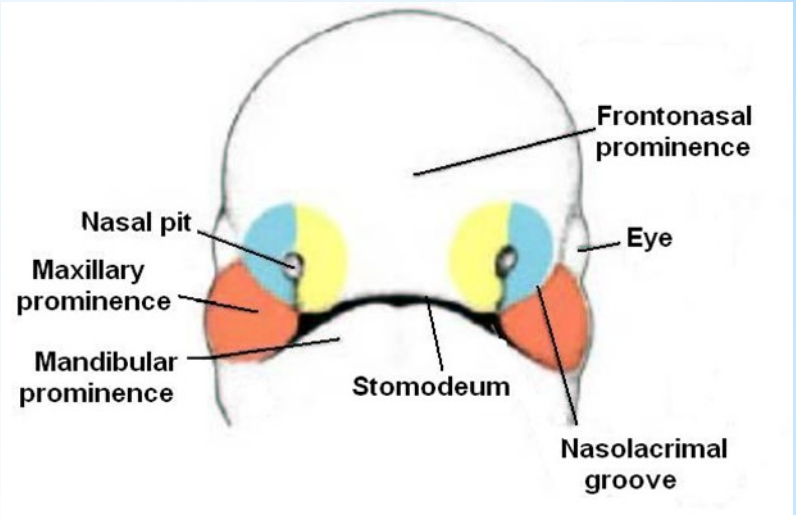


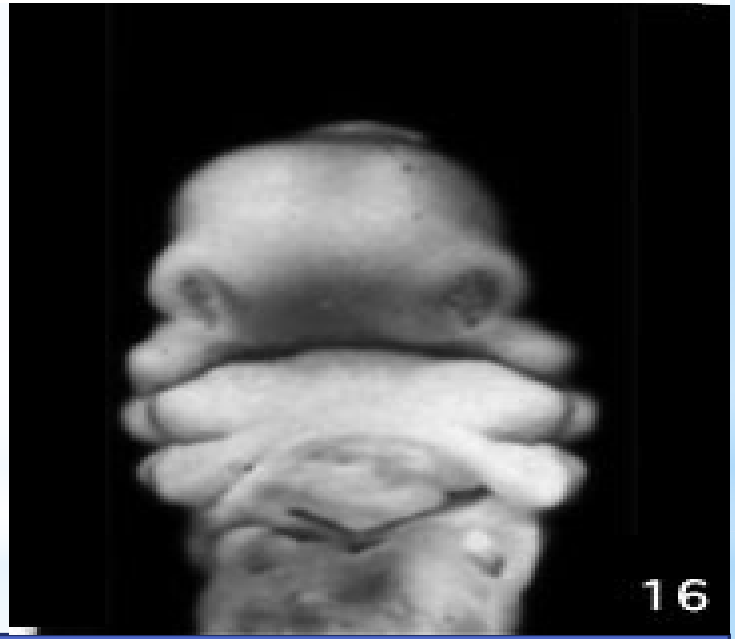
- Mesenchymal cells proliferate at the margin of the placodes and produce horse-shoe shaped swellings around these.
- The sides of these swellings are called ‘medial’ and ‘lateral’ nasal prominences.
- The placodes now lie in the floor of a depression called ‘nasal pits’.



Each lateral nasal prominence is separated from the maxillary swelling by nasolacrimal groove.

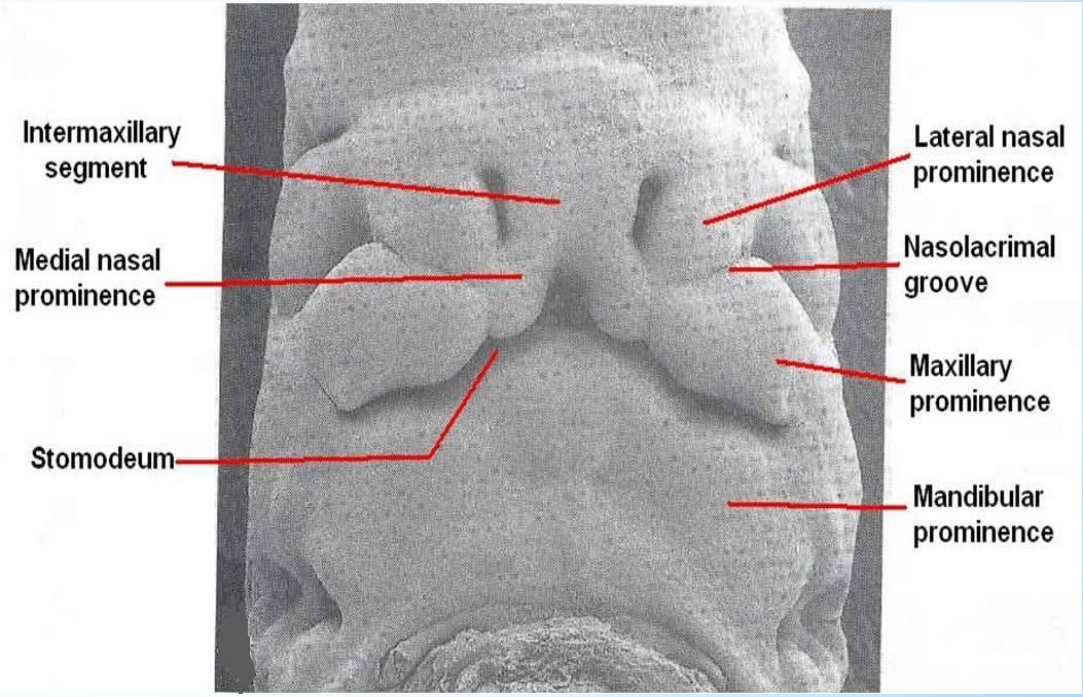
- The maxillary prominences continue to increase in size and:
- Laterally, merge with the mandibular prominences to form the cheek.
- Medially, compress the medial nasal prominences toward the midline and finally fuses with these to form the





The upper lip is formed by the two medial nasal prominences & the two maxillary prominences.

- The medial nasal swellings enlarge, grow medially and merge with each other in the midline to form the intermaxillary segment.
  - Gives rise to the:
    - Philtrum of lip.
    - Premaxillary part of the maxilla, that bears the upper 4 incisors and the associated gums.
    - Primary palate.



DERIVATIVES OF FACIAL COMPONENT

- The frontonasal prominence forms the:
  - Forehead and the bridge of the nose.
  - Frontal and nasal bones.

- The maxillary prominences form the:
  - Upper cheek regions and most of the upper lip.
  - Maxilla, zygomatic bone, secondary palate.

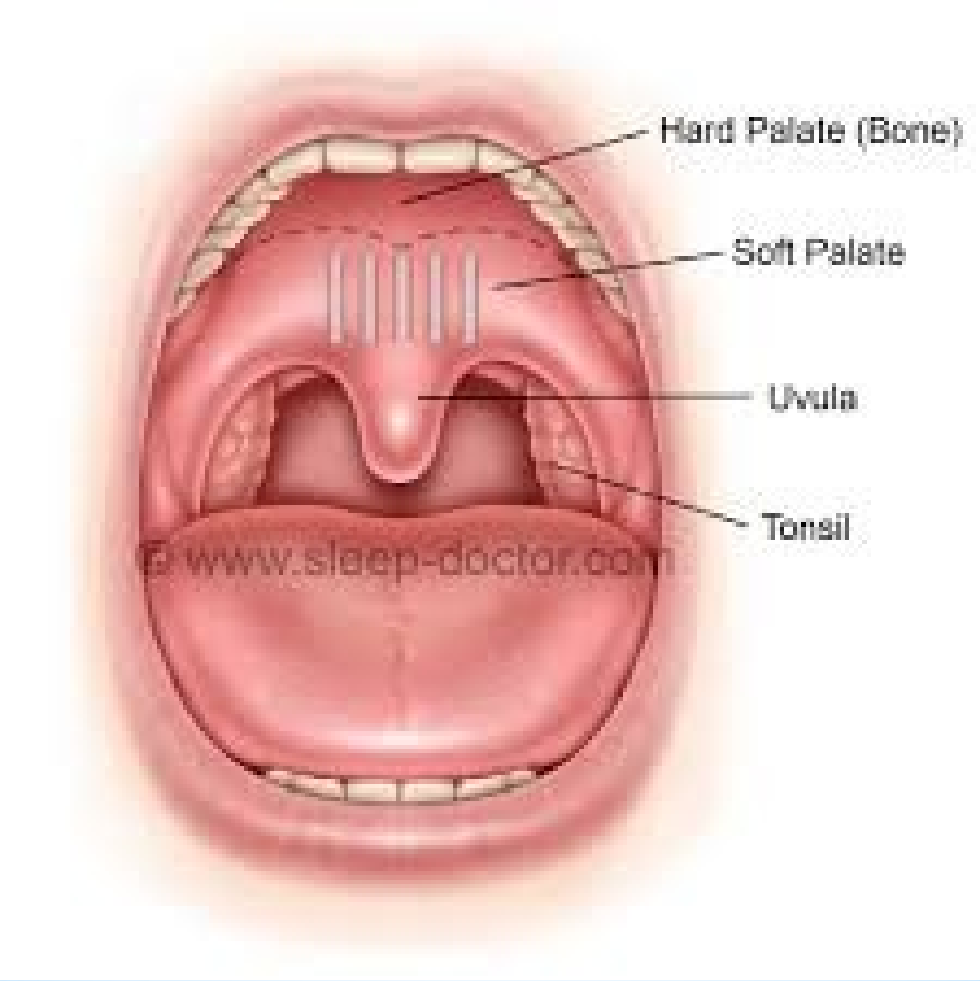
- The mandibular prominences fuse and form the:
  - Chin, lower lip, and lower cheek regions.
  - Mandible.

- The lateral nasal prominences form the alae

  of the nose.

- The medial nasal prominences fuse and form the intermaxillary segment.

Development of Palate (Palatogenesis)



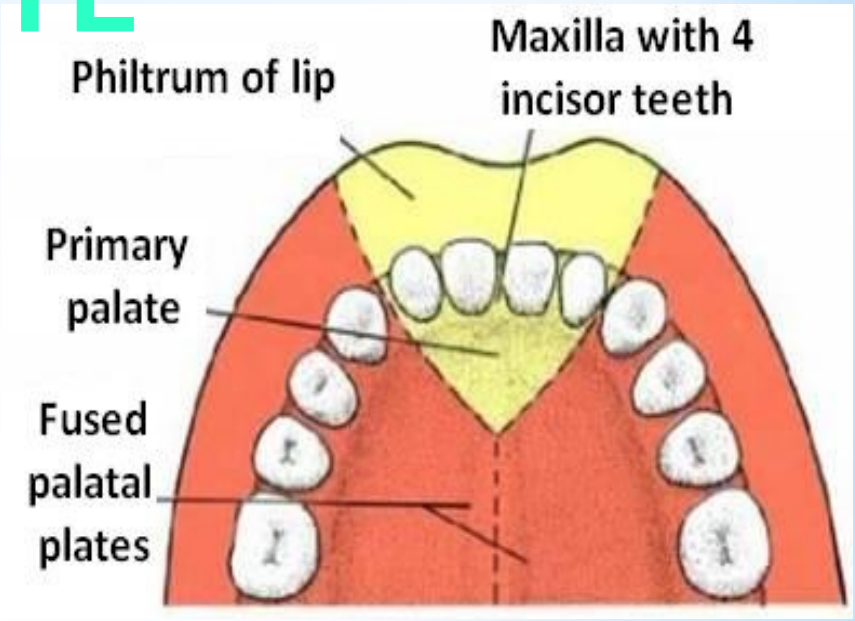
PALATOGENESIS

- Begins at the end of the 5th week.
- Gets completed by the end of the 12th week.

The palate develops from two primordia:
1: The primary palate
2: The secondary palate

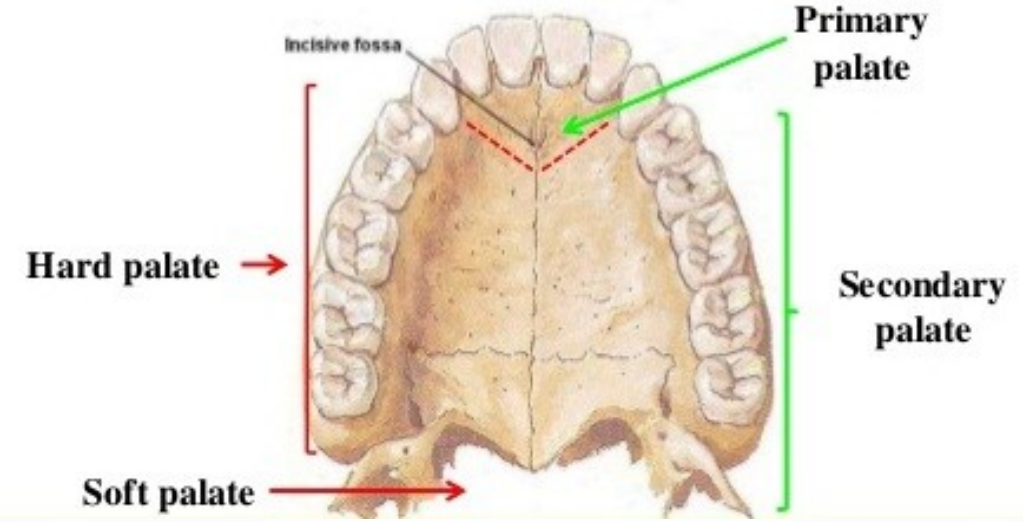
THE PRIMARY PALATE

- Begins to develop:
  - Early in the 6th week
  - From the deep part of the intermaxillary segment, as median palatine process.



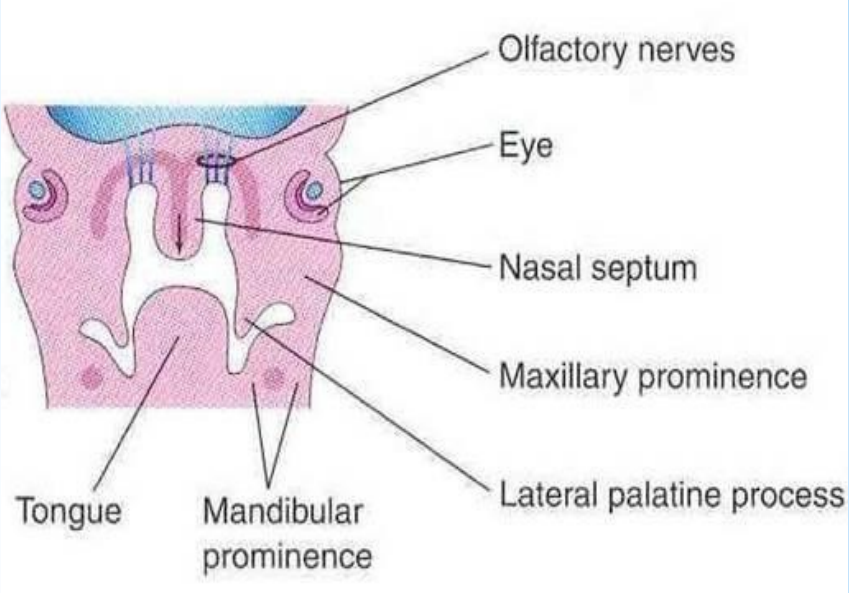
- Lies behind the premaxillary part of the maxilla.
- Fuses with the developing secondary palate.

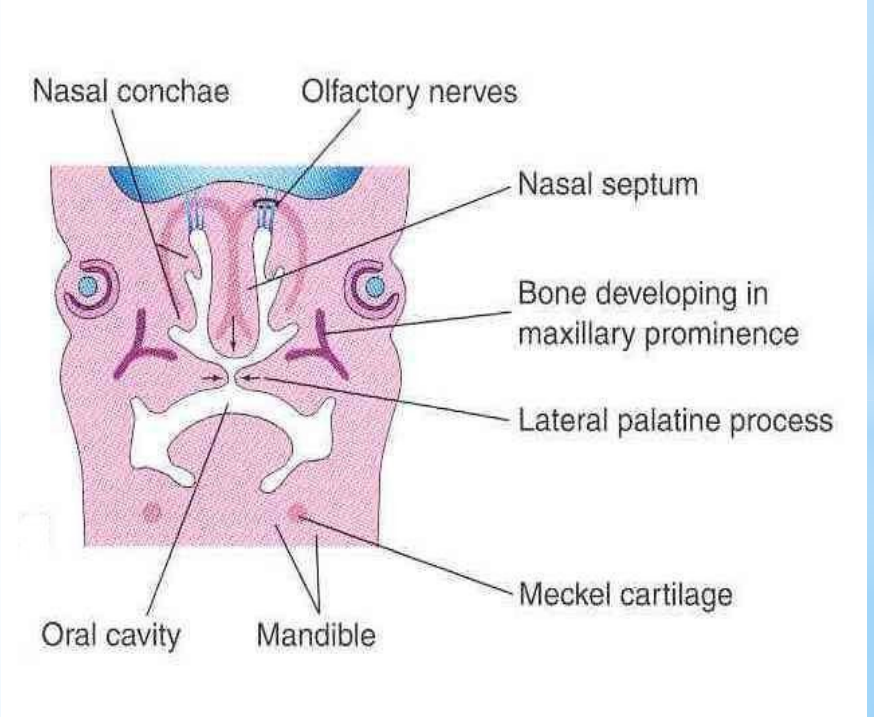
THE SECONDARY PALATE



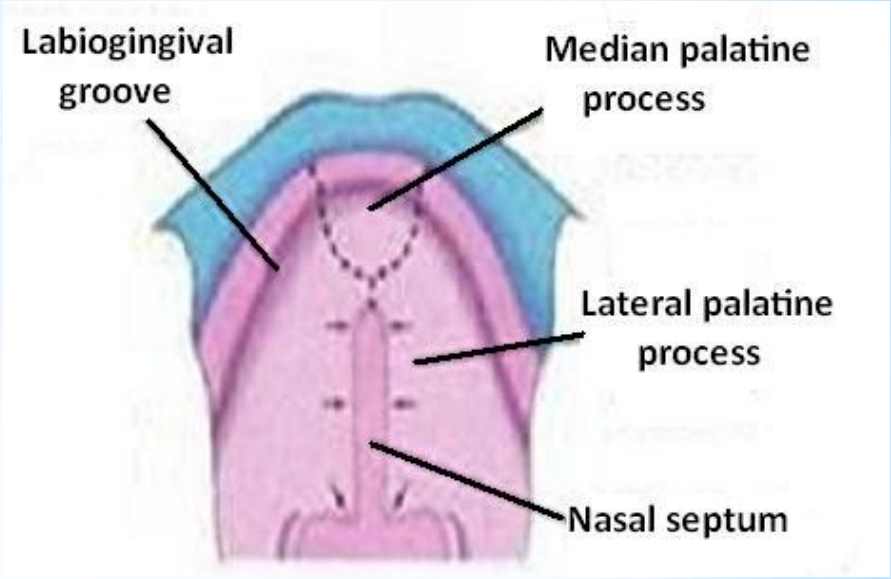
- It is the primordia of hard and soft palate posterior to the incisive fossa.
- Begins to develop:
  - Early in the 6th week.

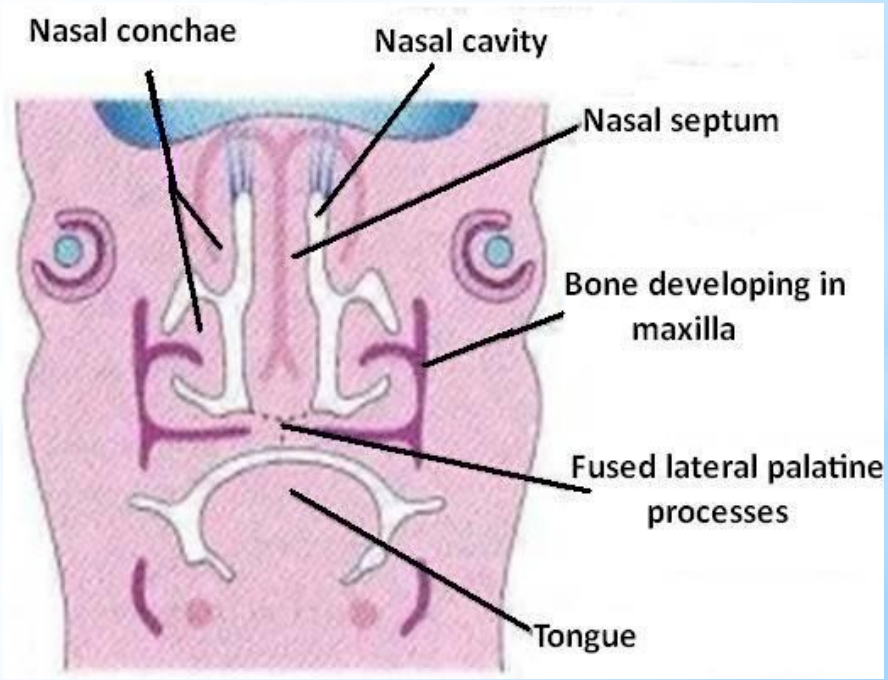
- In the beginning, the lateral palatine processes project inferomedially on each side of the tongue.
- With the development of the jaws, the tongue moves inferiorly.
- During 7th & 8th weeks, the lateral palatine processes elongate and ascend to a horizontal position above the tongue.



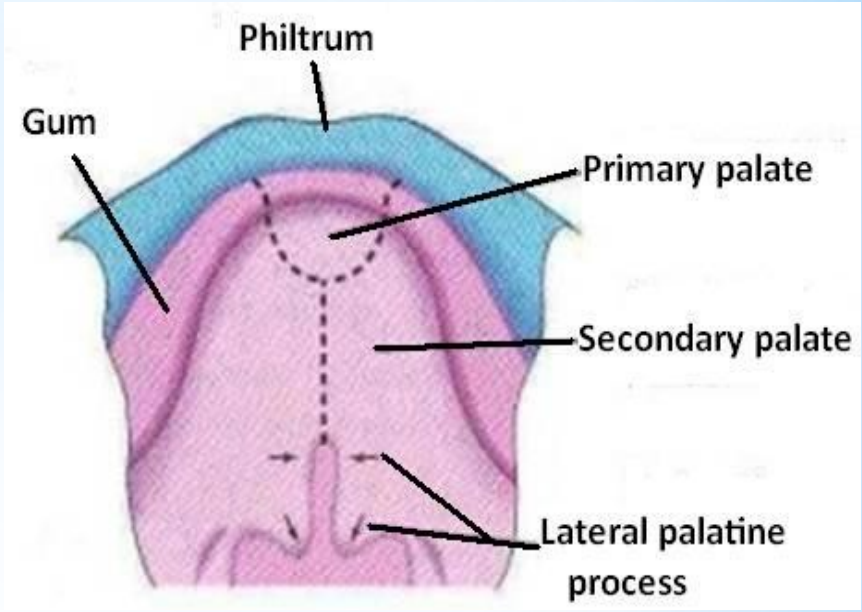


- Gradually the lateral palatine processes:
  - Grow medially and fuse in the median plane.
  - Also fuse with the: Posterior part of the primary palate & nasal septum.

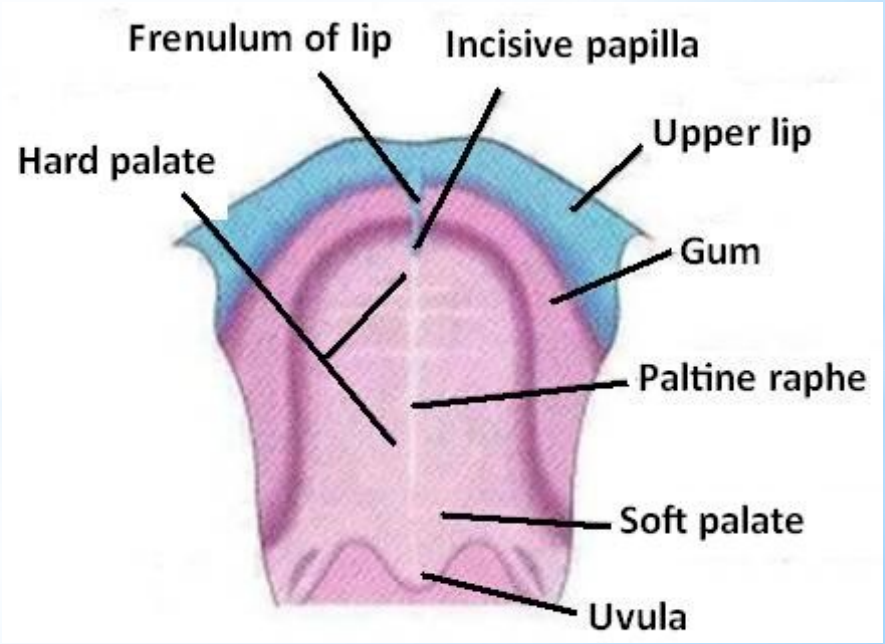




- Fusion with the nasal septum begins anteriorly during 9th week, extends posteriorly and is completed by 12th week.



Bone develops in the anterior part to form the hard palate. The posterior part develops as muscular soft palate.



Summary: Structures Contributing to Formation of the Face

Prominence	Structures Formed
Frontonasal	- One in number - Forms: Forehead, bridge of nose, and medial and lateral nasal prominences
Maxillary (Two)	- One on each side. Merge with the lateral nasal process then with the medial nasal process to form: - Part of Cheeks, Upper lip except the median part (Philtrum); Upper Jaw except median part; Definitive palate
Two Medial nasal (from frontonasal Process)	-- Two arise from lower border of frontonasal process; one on each side medially. - Merge with each other in the middle line to form intermaxillary segment which will give: - Philtrum of upper lip; Median part of upper jaw (carrying 4 incisors); Primitive palate; Primitive nasal septum; crest& tip of the nose
Two Lateral nasal (from frontonasal process)	-Two arise from lower border of frontonasal process; one on each side laterally. Separated from the maxillary process by cleft where the nasolacrimal duct lies - Give: Alae of nose
Mandibular (two)	- One on each side below the maxillary processes. Merge with its fellow of other side to form: - Whole Lower lip; Whole lower jaw; Most of cheek

SUMMARY OF DEVELOPMENT OF THE PALATE

I. Stage of formation of processes:

1- Formation of Primitive or Primary Palate

- By 5 th week: Intermaxillary process of frontonasal process→ 1ry palate→ later becomes the premaxilla of hard palate→Incisive fossa. (the anterior triangular area bounded by the 4 incisor teeth).

2. Formation of Definitive (secondary) palate

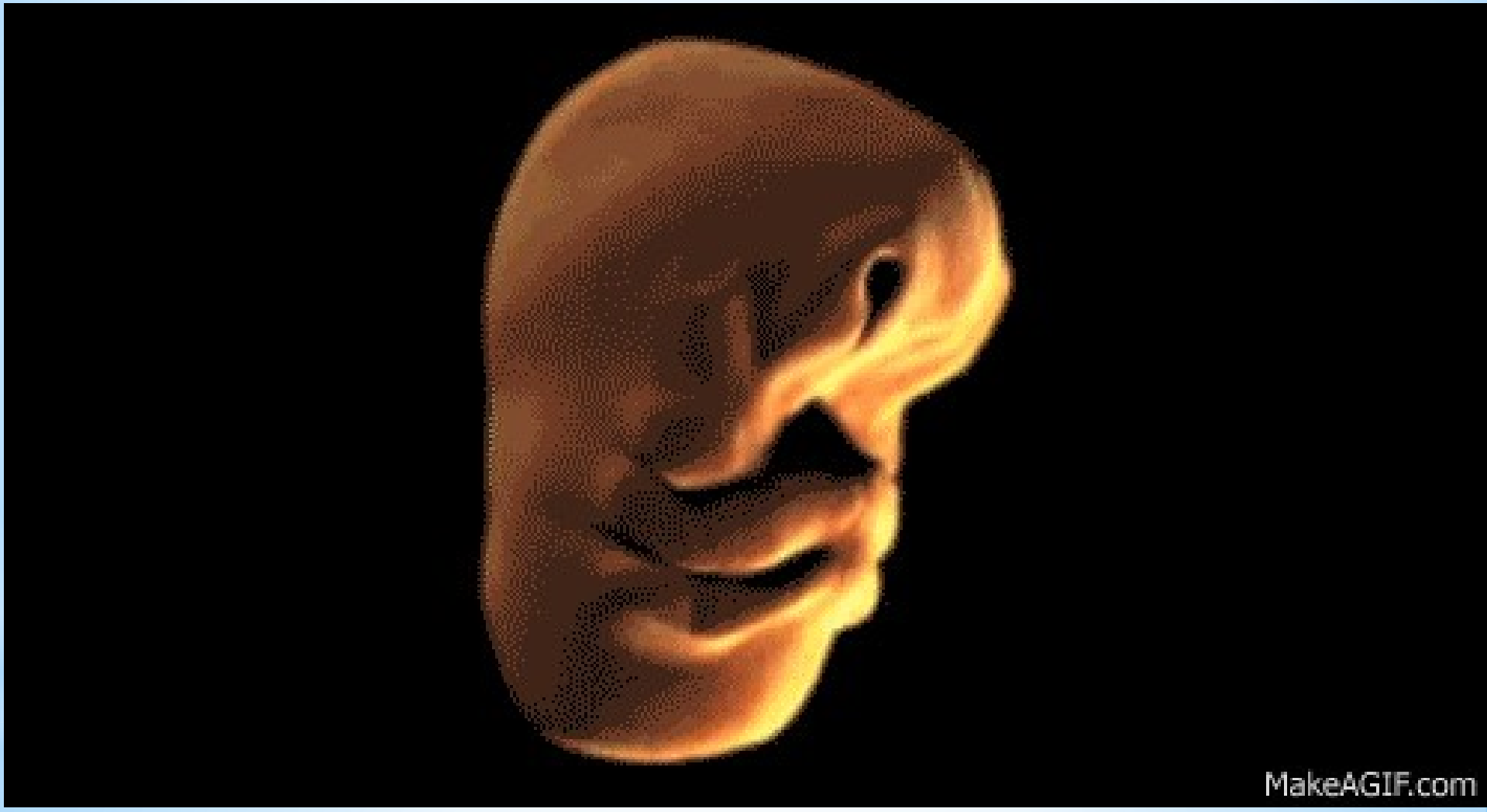
- Maxillary processes → palatine processes→ 2ry palate → meeting& fusion with 1ry palate.

II. Stage of meeting& fusion of the processes:

- Palatine processes grow medially& fuse with each other in the midline & fuse with the primitive palate. Fusion of the premaxilla, palatine processes of maxilla& nasal septum begins at 8 th week

III- Stage of ossification to form Hard palate:

- The anterior ¾ of the palate→ ossify→ hard palate & posterior ¼ → the soft palate & uvula.


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An ugly
personality
destroys
a pretty
face.